

LAWRENCE, MA, USA

WMO: 744904

Lat: 42.717N

Lon: 71.124W

Elev: 45

StdP: 100.78

Time Zone: -5.00 (NAE)

Period: 97-19

WBAN: 94723

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-15.2	-12.5	-24.3	0.4	-12.9	-22.0	0.5	-10.6	10.9	-3.1	9.7	-2.9	3.9	300	0.503

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	10.3	32.7	22.8	31.2	22.1	29.4	21.2	24.4	30.0	23.6	28.6	22.7	27.4	4.5	250

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
22.6	17.4	26.9	22.1	16.9	26.5	21.2	15.9	25.6	73.6	30.0	70.3	28.6	67.1	27.4	27.2

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
9.2	8.2	7.3	-18.3	35.3	2.9	1.7	-20.4	36.5	-22.1	37.5	-23.8	38.4	-25.9	39.6		
			-19.1	26.0	2.7	0.7	-21.1	26.5	-22.7	26.9	-24.3	27.3	-26.2	27.8		
			DB													
			WB													

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	10.4	-2.4	-1.3	2.7	9.1	14.7	19.5	22.9	22.1	18.1	11.5	5.9	0.6	(6)
(7)		DBStd	9.91	5.86	5.19	5.25	4.63	4.32	4.03	3.19	3.12	3.99	4.33	4.69	4.99	(7)
(8)		HDD10.0	1478	385	316	235	71	6	0	0	0	1	32	139	293	(8)
(9)		HDD18.3	3319	643	548	486	281	132	33	3	5	52	215	373	549	(9)
(10)		CDD10.0	1607	1	1	8	43	151	284	400	376	244	80	17	2	(10)
(11)		CDD18.3	407	0	0	1	3	18	68	145	123	46	4	0	0	(11)
(12)		CDH23.3	3522	0	1	9	47	216	612	1285	989	333	31	0	0	(12)
(13)		CDH26.7	1176	0	0	2	14	68	206	467	320	96	2	0	0	(13)
(14)	Wind	WSAvg	3.1	3.5	3.6	3.8	3.6	3.1	2.9	2.7	2.5	2.6	3.0	3.2	3.3	(14)
(15)	Precipitation	PrecAvg	1083	92	77	95	96	92	90	82	78	95	96	102	94	(15)
(16)		PrecMax	1539	328	218	288	307	200	307	180	232	234	371	246	218	(16)
(17)		PrecMin	716	14	13	12	18	22	12	17	17	17	11	20	0	(17)
(18)		PrecStd	175	71	40	52	52	44	66	42	46	57	68	52	51	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	16.3	16.2	23.7	28.7	31.5	33.6	34.8	34.1	32.7	26.4	21.0	17.6	(19)
MCWB			13.3	11.4	15.3	17.0	19.8	23.3	23.9	23.7	22.8	18.6	15.6	14.3	(20)	
2%		DB	11.7	11.7	17.5	23.4	28.5	31.3	32.7	32.1	29.2	23.7	18.0	13.0	(21)	
		MCWB	8.3	7.8	11.4	14.3	18.9	22.5	23.1	22.7	21.2	18.1	14.0	10.2	(22)	
5%		DB	8.3	8.6	13.3	20.2	25.7	29.0	31.1	30.0	27.0	21.2	16.0	10.5	(23)	
		MCWB	5.5	5.0	8.1	12.6	17.1	20.7	22.4	21.7	20.2	16.8	12.0	8.0	(24)	
10%		DB	5.4	6.0	10.2	17.3	22.8	27.1	29.1	28.2	24.8	18.7	13.3	7.9	(25)	
		MCWB	2.6	2.7	6.2	11.0	15.9	19.8	21.6	21.0	19.2	14.7	10.3	5.4	(26)	
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	14.0	13.6	16.2	18.5	21.7	24.9	25.7	25.2	24.0	21.3	17.5	14.6	(27)
MCD B			15.5	15.6	22.2	24.6	29.5	31.2	32.5	31.7	29.4	24.2	19.9	16.7	(28)	
2%		WB	8.8	8.7	12.6	16.1	20.2	23.6	24.5	24.1	22.7	19.1	15.2	11.1	(29)	
		MCD B	10.8	11.2	16.6	22.4	26.3	29.1	30.3	29.2	27.1	22.3	17.6	12.8	(30)	
5%		WB	5.6	5.2	9.0	14.0	18.6	22.4	23.6	23.3	21.7	17.2	12.8	8.3	(31)	
		MCD B	8.4	8.0	12.0	18.8	23.7	27.2	28.9	27.7	25.3	20.6	15.0	9.9	(32)	
10%		WB	2.9	3.1	6.5	11.4	16.9	21.1	22.7	22.4	20.6	15.3	10.6	5.8	(33)	
		MCD B	5.0	5.1	9.8	16.1	21.4	25.3	27.6	26.5	23.9	18.0	13.1	7.8	(34)	
(35)	Mean Daily Temperature Range	MDBR		8.0	8.9	9.1	10.5	10.7	10.3	10.3	10.2	10.4	9.7	8.7	7.8	(35)
5% DB		MCD BR	11.2	11.9	13.5	15.5	15.3	13.0	12.2	12.0	12.2	12.3	11.6	10.8	(36)	
		MCW BR	9.5	9.0	8.8	8.5	7.5	6.1	5.0	4.9	6.0	7.5	8.7	9.4	(37)	
5% WB		MCD BR	11.1	11.0	12.5	13.9	13.5	11.5	10.6	10.1	9.8	11.1	9.9	10.6	(38)	
		MCW BR	9.9	9.2	9.5	9.2	7.5	6.2	5.2	5.0	5.9	8.0	8.6	10.1	(39)	
(40)	Clear-Sky Solar Irradiance	taub		0.285	0.295	0.320	0.363	0.399	0.441	0.461	0.428	0.371	0.331	0.307	0.289	(40)
taud		2.489	2.479	2.484	2.400	2.338	2.249	2.203	2.277	2.421	2.542	2.568	2.561	(41)		
Ebn at Noon		866	914	928	905	875	836	814	830	859	861	836	833	(42)		
Edh at Noon		71	85	96	113	124	135	141	126	102	80	67	62	(43)		
(44)	All-Sky Solar Radiation	RadAvg		1.75	2.61	3.63	4.63	5.14	5.55	5.86	5.28	4.25	2.78	1.89	1.43	(44)
(45)		RadStd		0.11	0.21	0.24	0.50	0.59	0.52	0.37	0.24	0.28	0.25	0.16	0.15	(45)

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days					
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)		
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+137	+85		
(47) Regional (75 neighbors)		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+91	+38		

Nomenclature: See separate page